Foreign Currency Exchange Rate Forecast: Replicating CS229 #34 on Public Data

Course: Machine Learning — Mini Project

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Abstract

We replicate the Stanford CS229 project *Foreign Currency Exchange Rate Forecast* (Zhang, Sun & Zhang, 2019). The study forecasts daily movements of GBP, EUR and JPY against the US dollar with six machine-learning models, in two experiments: next-day direction (classification) and next-day log return (regression), the second followed by a $100 trading simulation. We use the paper's currency pairs, features, models, 80/10/10 split, metrics and trading rule. Bloomberg and CRSP are replaced by free Yahoo Finance and FRED data covering December 2003 – November 2018. We add explicit baselines, point-in-time alignment of economic data, removal of bad price records, transaction costs and significance tests. No model shows reliable skill. Test-set direction accuracy ranges from 0.503 to 0.529 against a coin flip of 0.5, and none is significant after correcting for the number of models tried. Every regression model's test R^2 is between −0.052 and +0.004 relative to a random walk. The model chosen on the validation set is Lasso with every coefficient shrunk to zero, i.e. a constant forecast. The paper's Random Forest results (test R^2 0.70–0.78, up to $110.50 from $100) do not replicate: we obtain R^2 between −0.06 and −0.02 and $83–$97 after costs. Our results agree with Meese & Rogoff (1983): past prices and public macro data do not predict daily exchange-rate moves.

1. Introduction

Exchange rates set the price of every cross-border transaction, so firms, investors and governments face currency risk. Forecasting them is notoriously hard. Meese and Rogoff (1983) showed that a random walk ("tomorrow's rate equals today's") forecasts better than structural economic models, and the result has largely held for four decades.

The CS229 project #34 asked whether modern machine learning changes that. It reported that a Random Forest explains about 74% of the variance of daily log returns and earns up to 10.5% in a simple trading simulation over 2017–2018.

Our goal is to reproduce that study as closely as possible on publicly available data, and to test its claims against explicit baselines with realistic data timing. Our contributions:

1. A faithful, open re-implementation of both experiments and the profit simulation.
2. Fixes needed for a valid result: random-walk and majority baselines, economic data used only after its publication date, removal of isolated bad price records, and trading costs.
3. Significance testing, plus validation-period and random-seed checks for every result that looked better than chance.

2. Related Work

- **Zhang, Sun & Zhang (2019), CS229 #34.** Daily USD/GBP, USD/EUR and USD/JPY closes, 2001–2018 (Bloomberg). They use 15 engineered exchange-rate features plus stock-index and US macro factors (CRSP), an 80/10/10 split, and Logistic, Lasso, Ridge, SVM, MLP and Random Forest models. Results: sign accuracy about 0.51–0.55 (their Table 1), Random Forest selected by validation R^2 of 0.74 (Table 2), test R^2 0.70–0.78 (Table 4), and final cash \$100.33–\$110.50 (Table 3).
- **Meese & Rogoff (1983).** The random walk beats structural exchange-rate models out of sample.
- **Kuan & Liu (1995).** Neural networks for exchange rates; some show market-timing ability relative to the random walk.
- **Gu, Kelly & Xiu (2020).** Machine learning in empirical asset pricing. They use out-of-sample R^2 measured against a zero forecast, the convention we adopt (§7).
- **Campbell & Thompson (2008).** Out-of-sample R^2 relative to a naive benchmark as the standard test of return predictability.

3. Dataset

	Paper	Ours	Reason
Exchange rates	Bloomberg, daily close	Yahoo Finance `GBP=X`, `EUR=X`, `JPY=X` (foreign currency per USD)	Bloomberg is paid
Period	Jan 2001 – Nov/Dec 2018	**1 Dec 2003 – 30 Nov 2018**	Yahoo's GBP and EUR history starts Dec 2003
Stock indexes	S&P 500, S&P 100, Nasdaq, DJIA (CRSP)	same, Yahoo `^GSPC ^OEX ^IXIC ^DJI`	CRSP is paid
Economic factors	CPI change, Fed funds, NBER recession indicator, federal debt (CRSP)	same series from FRED: `CPIAUCSL`, `DFF`, `USREC`, `GFDEBTN`	not available on Yahoo

All raw files are committed under `data/raw/`. The end date matches the paper, so our test period (30 Jun 2017 – 29 Nov 2018) almost coincides with theirs (3 Mar 2017 – 30 Nov 2018).

Table 1 — Daily log returns after cleaning (`results/summary_stats.csv`)

Pair	Trading days	Rate range	Mean (%)	Std dev (%)	Skew	Excess kurtosis	Up days (%)
GBP	3,899	0.474 – 0.831	0.008	0.602	0.82	9.98	49.9
EUR	3,882	0.625 – 0.962	0.001	0.626	0.01	2.28	49.5
JPY	3,883	75.74 – 125.63	0.001	0.646	0.02	3.92	50.4

The mean daily return is essentially zero, the typical day moves about 0.6%, and up and down days are equally frequent. Before any model is trained, the data already suggest there is little to predict. GBP's high kurtosis comes mostly from the Brexit referendum move (+7.9% in one day).

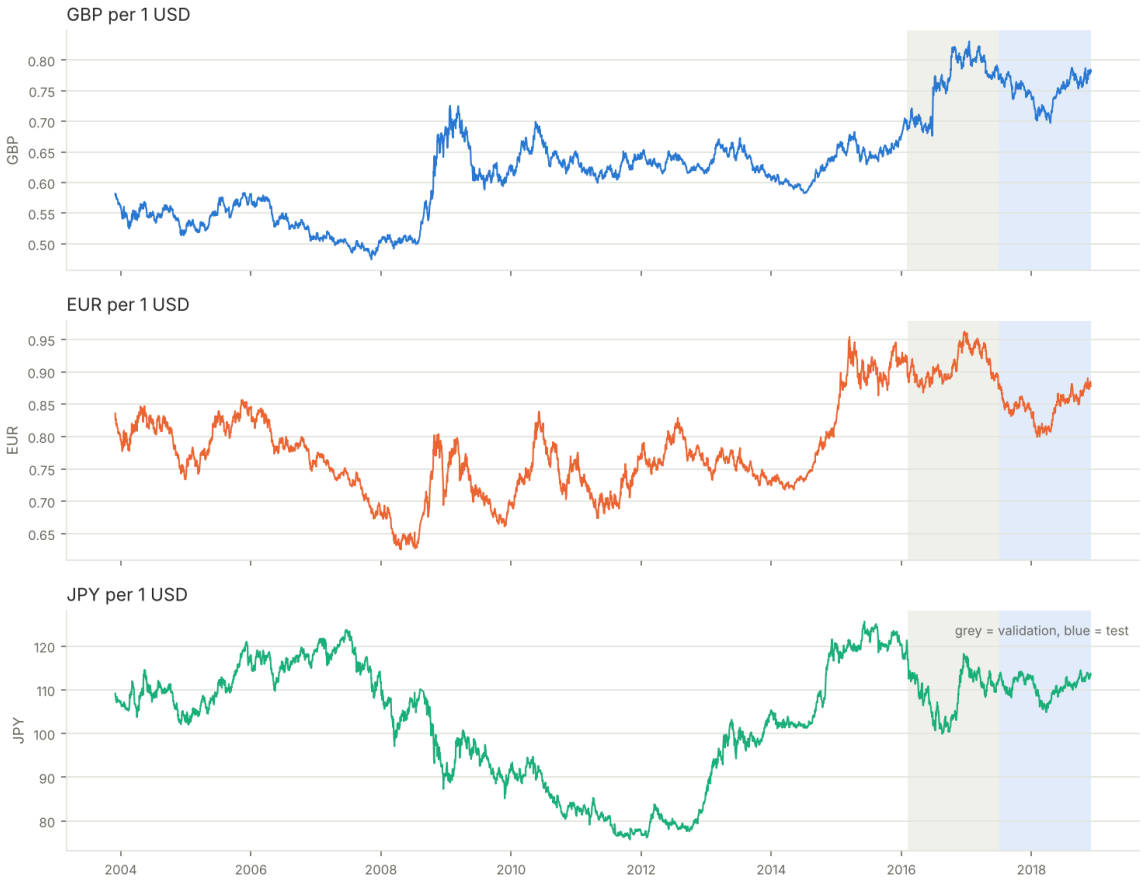


Figure 1. The three rates (foreign currency per USD); grey = validation, blue = test.

4. Preprocessing

From `results/data_report.json`:

- **Missing / duplicate / non-positive closes:** none in any series.
- **Weekends and holidays are not filled.** "Next day" means the next available trading day. Forward-filling would create fake 0% return days.
- **Calendar gaps:** EUR and JPY have no Yahoo rows for most of August 2008 (gaps ending 8 Aug and 26 Aug 2008); GBP has one gap ending 23 Jan 2008. These are left as gaps.
- **Point-in-time economic data.** Each value is attached to a trading day only from the date it would have been public: CPI and recession flag +45 days, federal debt +180 days, Fed funds +1 day, stock-index returns +1 trading day. Without this, March CPI (released mid-April) would be "known" on 1 March, which is look-ahead bias. A unit test enforces this.

Bad price records removed. Yahoo's EUR and JPY history contains isolated wrong prices in 2008. Each is a 3–15% one-day spike on the 8th of a month that fully reverses the next day. Example: EUR on 5 Sep 2008 = 0.7005, on 8 Sep = 0.6645, on 9 Sep = 0.7068. GBP shows nothing on the same dates.

Rule. Drop a row if its log price is more than 3% from the midpoint of its two neighbours, on the same side of both, while the neighbours agree with each other (a spike that snaps back). This removes **5 EUR rows** (8 Jan, 8 Feb, 8 Sep, 8 Oct, 8 Dec 2008) and **4 JPY rows** (8 Apr, 8 May, 8 Oct, 8 Dec 2008), and nothing in GBP. Genuine

moves such as Brexit are level shifts that do not snap back, so they are kept. The rule is applied to FX only; for stock indexes a 3% spike-and-revert is usually a real crisis day.

Effect. EUR excess kurtosis falls from 89.5 to 2.3, and JPY's from 110.9 to 3.9 (Figure 0). §9.4 shows our conclusions do not depend on this step.

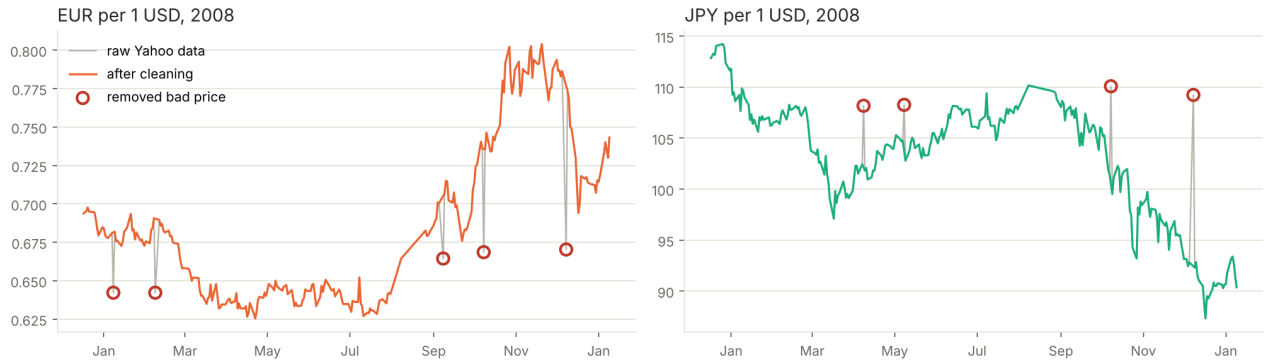


Figure 0. EUR and JPY in 2008: raw Yahoo data (grey), the cleaned series, and the removed bad prices (red circles).

5. Features

As in the paper (§3.2), built per pair for each trading day *t* using data up to *t* only:

Set	Features	Count
`fx7` (paper's initial set)	log return today, log return yesterday, mean log return over the last 5, 10, 20, 60, 120 days	7
`fx15` (paper's final set)	`fx7` + log rate, its square, its cube, mean log rate over the last 5, 10, 20, 60, 220 days	15
`fx15_macro` (paper's full set)	`fx15` + daily log returns of S&P 500, S&P 100, Nasdaq, DJIA + CPI % change, Fed funds rate, NBER recession flag, federal debt % change	23

Targets: Exp 1: `up` = 1 if tomorrow's log return > 0. Exp 2: tomorrow's log return r = $\log(\text{rate}_{t+1} / \text{rate}_t)$.

Leakage checks (`tests/test_pipeline.py`, 9 tests, all pass): changing every price after day *k* leaves all features up to *k* unchanged; macro values never appear before their publication lag; the split is strictly chronological.

The first usable row is 1 Oct 2004, because the 220-day average needs about a year of history.

6. Methods

Split (paper §3.1). Chronological 80 / 10 / 10 per pair:

Pair	Train	Validation	Test
GBP	2,943 days (Oct 2004 – 1 Feb 2016)	368 (Feb 2016 – Jun 2017)	368 (30 Jun 2017 – 29 Nov 2018)
EUR	2,929 (Oct 2004 – 4 Feb 2016)	366	367 (3 Jul 2017 – 29 Nov 2018)
JPY	2,930 (Oct 2004 – 4 Feb 2016)	366	367 (3 Jul 2017 – 29 Nov 2018)

Hyperparameters are chosen on validation; the chosen model is refit on train + validation and scored once on test.

Models (paper §4). All except Random Forest use standardised features.

Exp 1 (classification)	Exp 2 (regression)	Hyperparameters searched
Logistic regression	Linear regression (added, from our proposal)	—
Lasso = L1 logistic	Lasso	$C \in \{0.01, 0.1, 1\}$ / $\alpha \in \{0.001, 0.01, 0.1\}$
Ridge classifier	Ridge	$\alpha \in \{0.1, 1, 10, 100\}$
SVM, RBF kernel	SVR, RBF and polynomial (degree 3) kernels	$C \in \{0.1, 1, 10\}$ / $\{0.1, 1\}$
Random Forest, 1000 trees, max depth 100 (as in the paper)	same	—
MLP (64, 32), early stopping	same	$\alpha \in \{10^{-4}, 10^{-2}\}$

In Exp 2 the target is standardised too. Daily returns are around 0.005, far below SVR's default $\varepsilon = 0.1$, so without this SVR would treat every error as zero.

Baselines (added). Exp 1: *majority class* and *same direction as today*. Exp 2: *random walk* (predict 0 return, i.e. tomorrow's rate = today's), *historical mean* return, and *momentum* (tomorrow's return = today's).

7. Experiments and Metrics

Exp 1, sign prediction: the paper's metrics, with up = +1 and down = −1. Acc = share correct; NMSE = mean$((s - \hat{s})^2)$ / var(s); MAE = mean$|s - \hat{s}| = 2(1 - \text{Acc})$; DS = share with matching direction (= Acc for a classifier); CU = share of actual up days predicted up; CD = share of actual down days predicted down.

Exp 2, log-return regression:

- R^2, the paper's metric, used on validation to choose the model as the paper did.
- R^2 vs random walk = 1 − SSE(model) / SSE(predict 0). Above 0 means the model beats "tomorrow = today" (Campbell & Thompson, 2008; Gu, Kelly & Xiu, 2020).
- MAE and RMSE in basis points (1 bp = 0.01%), and directional accuracy.

Profit simulation (paper §5.2). Start with \$100. Each day, take the predicted ratio $\hat{R} = e^{\hat{r}}$ for each currency. If all $\hat{R} \geq 1$, stay in USD. Otherwise hold the currency with the lowest $\hat{R}$ for one day; cash is multiplied by 1 / (actual ratio). This is run for all 7 currency combinations at **0 cost (paper)** and **2 bp per conversion (added)**. A test reproduces the paper's worked example (\$100 / 0.95 = \$105.26).

Significance (added). For each model, test days are pooled across the three pairs, and a one-sided binomial test asks whether directional accuracy beats 0.5. Bonferroni correction is applied for the number of models tried (18 in Exp 1, 14 in Exp 2).

8. Results

8.1 Experiment 1 — sign prediction (paper Table 1)

Table 2 — Test set, paper's full feature set (`fx15_macro`), averaged over the 3 pairs. Paper values in brackets.

Model	Acc	NMSE	MAE	DS	CU	CD
Logistic	0.505 (0.51)	1.980 (1.97)	0.989 (0.98)	0.505 (0.51)	0.570 (0.45)	0.439 (0.57)
Lasso	0.505 (0.52)	1.983 (1.90)	0.991 (0.95)	0.505 (0.52)	0.526 (0.47)	0.481 (0.58)
Ridge	0.518 (0.55)	1.929 (1.81)	0.964 (0.91)	0.518 (0.55)	0.509 (0.58)	0.523 (0.51)
SVM (RBF)	0.518 (0.51)	1.929 (1.95)	0.964 (0.97)	0.518 (0.51)	0.462 (0.57)	0.570 (0.52)
Random Forest	0.505 (0.53)	1.983 (2.03)	0.991 (1.02)	0.505 (0.49)	0.623 (0.44)	0.393 (0.49)
MLP	0.529 (0.52)	1.885 (1.92)	0.942 (0.96)	0.529 (0.52)	0.540 (0.54)	0.514 (0.52)
Majority class	0.496	2.016	1.007	0.496	0.667	0.333
Same as today	0.476	2.096	1.047	0.476	0.465	0.487

All 18 model × feature-set combinations score between **0.503 and 0.529** (`results/exp1_sign_avg.csv` , Figure 2). This **agrees with the paper**, whose models scored 0.51–0.55. Our encoding also reproduces the paper's internal arithmetic: their Lasso has Acc 0.52 and MAE 0.95, and 2 × (1 − 0.52) = 0.96.

Is anything better than chance? (`results/significance_exp1.csv`)

- The best test accuracy, MLP on `fx15_macro` (0.529, 583/1102 days), has one-sided p = 0.029, but **p = 0.52 after Bonferroni** correction for 18 models.
- The models validation would have chosen score 0.518 (SVM, `fx15_macro`), 0.511 (MLP, `fx7`) and 0.505 (RF, `fx15`). None is significant: p = 0.12, 0.24 and 0.37.

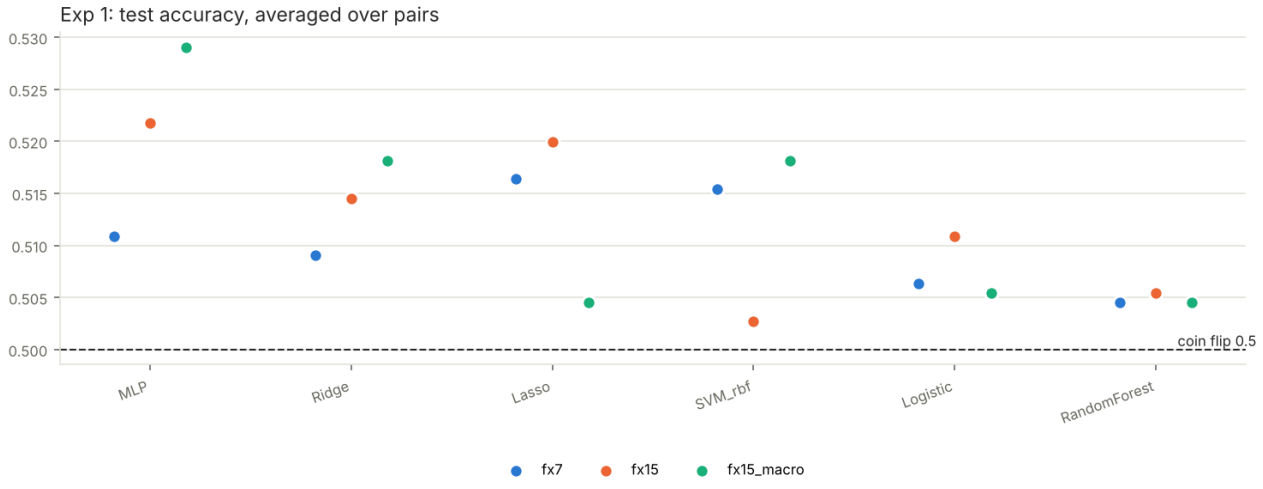


Figure 2. Exp 1 test accuracy for every model and feature set; dashed line = coin flip.

8.2 Experiment 2 — log-return regression (paper Tables 2 and 4)

Table 3 — Validation R^2 used for model selection (paper Table 2), `fx15_macro` , mean over pairs

	Random Forest	MLP	SVM RBF	SVM poly	Lasso	Ridge	Linear reg.
Paper	**0.74**	−∞	0	−0.02	0.01	0.01	—
Ours	−0.667	−0.044	−0.007	−0.181	**−0.0005**	−0.028	−0.806

The paper selected Random Forest. With the same procedure we select **Lasso**: validation picks α = 0.1, which shrinks every coefficient to zero. The "best" model therefore ignores all 23 features and predicts the constant training-set mean return every day.

Table 4 — Test set, mean over pairs (`results/exp2_return_avg.csv`)

Model	Features	R^2 vs random walk	MAE (bp)	RMSE (bp)	Direction acc.
Random walk	—	0 (by definition)	36.38	46.66	—
Historical mean	—	−0.00001	36.39	46.66	0.489
Momentum	—	−1.112	54.48	67.76	0.477
Lasso (selected)	fx15 / fx15_macro	−0.00001	36.39	46.66	0.489
Linear regression	fx15_macro	−0.014	36.70	47.01	0.503
Ridge	fx15_macro	−0.013	36.58	46.98	0.504
SVM RBF	fx15_macro	−0.011	36.54	46.94	0.513
SVM poly	fx15_macro	+0.004	36.27	46.57	0.554
Random Forest	fx15_macro	−0.034	37.17	47.46	0.489
Random Forest	fx15	−0.052	37.50	47.88	0.487
MLP	fx15_macro	−0.031	36.79	47.40	0.535

Table 5 — Random Forest test R^2 by pair (paper Table 4)

	GBP	EUR	JPY
Paper	0.70	0.78	0.74
Ours (`fx15_macro`)	−0.062	−0.021	−0.018

Only one model is above zero: SVM poly at +0.004, i.e. 0.4% lower squared error than the random walk. Random Forest is the worst learned model, 3.4–5.2% worse than doing nothing.

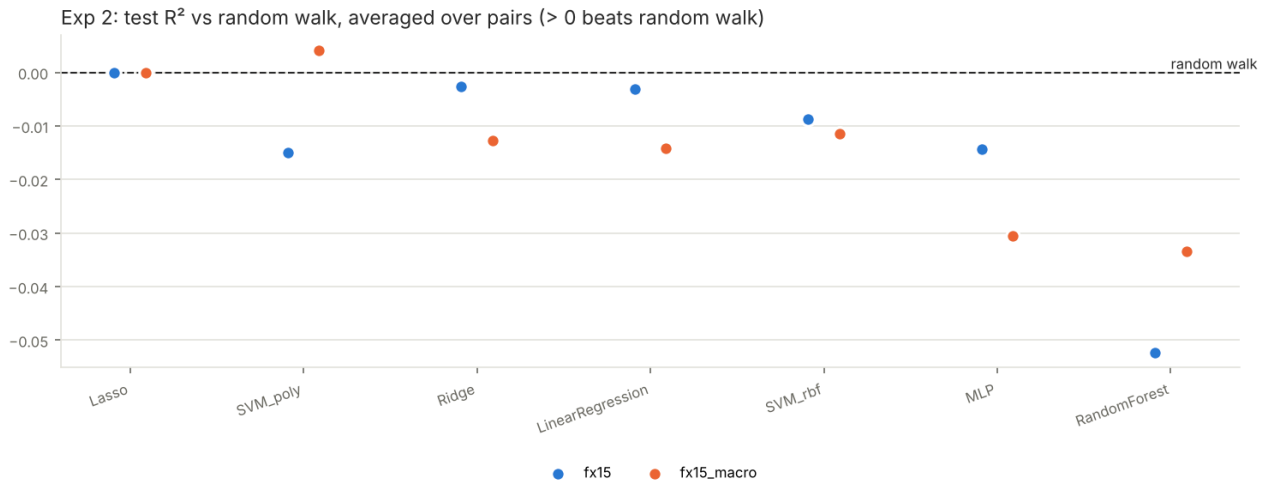


Figure 3. Test R^2 relative to the random walk; above the dashed line = better than "tomorrow = today".

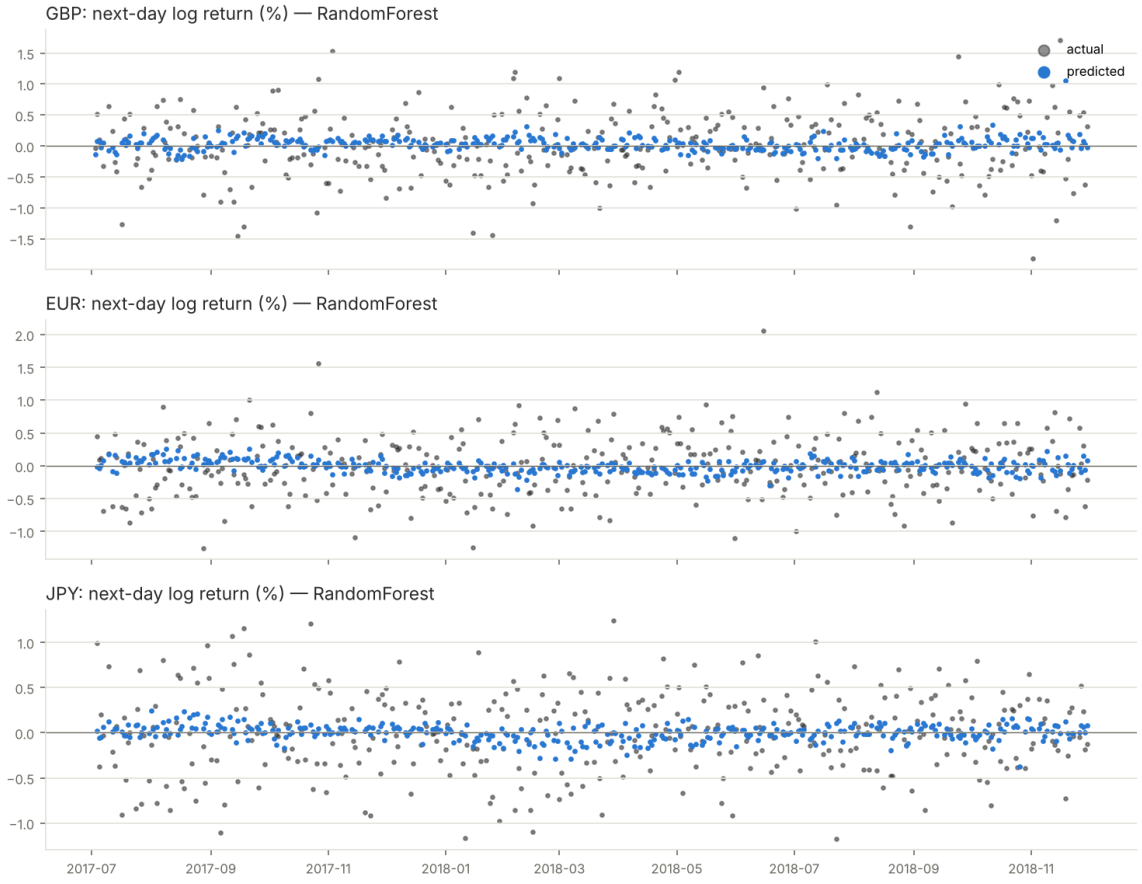


Figure 4. Random Forest predicted vs actual next-day log returns (paper Figure 2). Predictions cluster near zero while actual returns spread ±1.5%.

Two results that looked like skill, and why they are not

1. **SVM poly direction accuracy 0.554** (p = 0.0002; Bonferroni p = 0.002). Validation chose neither this model nor its hyperparameters (validation R^2 −0.18). Trained on the training set only, its directional accuracy on the *validation* period is 0.489 (GBP), 0.481 (EUR) and 0.514 (JPY), i.e. no skill. A real signal should appear in both periods; this one appears only in the 17-month test window (`results/diagnose_SVM_poly_fx15_macro.csv`).

2. **MLP direction accuracy 0.535** (p = 0.010; Bonferroni p = 0.14). Re-training with five other random seeds gives 0.474, 0.516, 0.492, 0.502 and 0.499. The default seed (42) happened to be the luckiest (`results/seeds_MLP_fx15_macro.csv`).

8.3 Profit simulation (paper Table 3, Figure 3)

Table 6 — Final cash from $100, Random Forest, test period (`results/backtest_summary.csv`)

Currencies traded	Paper	Ours, no cost	Ours, 2 bp/side	Momentum, no cost	Days held foreign
GBP	104.40	93.81	88.52	98.26	39%
JPY	104.44	103.03	96.61	93.22	44%
EUR	110.50	100.75	93.12	97.34	54%
GBP + JPY	100.33	95.82	86.73	89.19	68%
JPY + EUR	109.46	97.56	87.78	89.00	72%
GBP + EUR	104.88	97.09	87.57	95.07	70%
GBP + JPY + EUR	101.30	93.94	83.15	89.42	83%

- Without costs, the Random Forest loses money in 5 of 7 cases. With all three currencies it peaks at $106.91 (Feb 2018), then falls to $93.94 (max drawdown −13.1%).
- With a 2 bp cost per conversion, **all 7 cases lose money ($83–$97)**. Trading on 39–83% of days makes costs dominate.
- The selected model (Lasso) predicts a small positive return every day, so $\hat{R} > 1$ for every currency and it never trades: **$100.00 in all cases**.

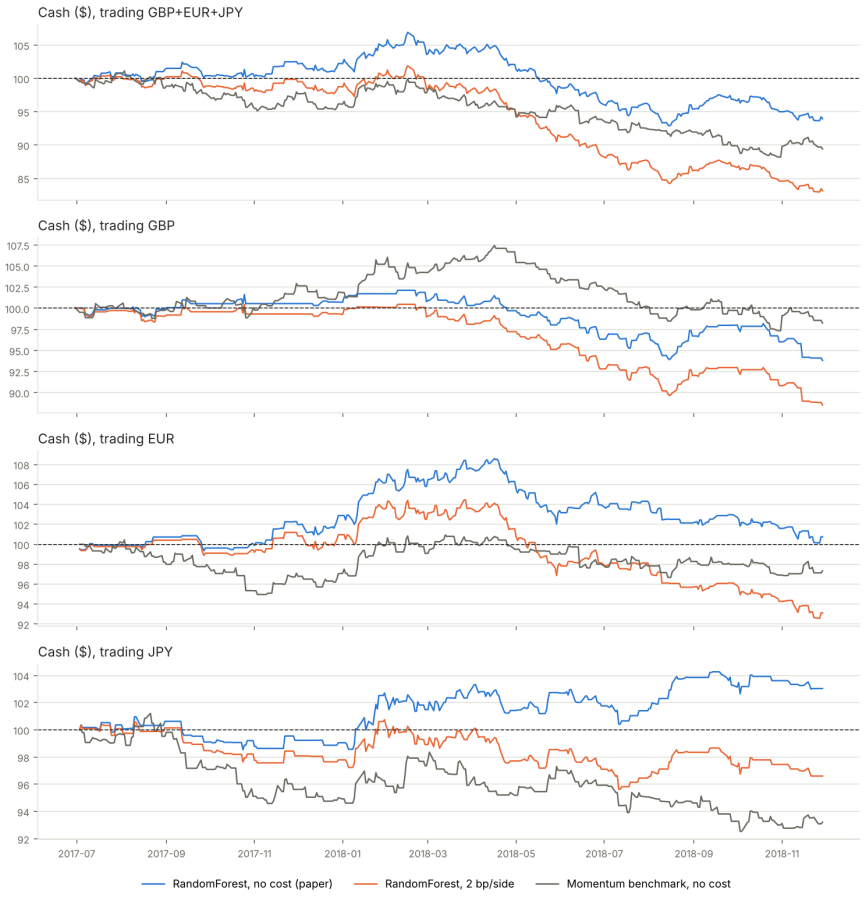


Figure 5. Cash over the test period: Random Forest without cost (paper setting), with 2 bp per side, and the momentum benchmark.

8.4 What the Random Forest relied on

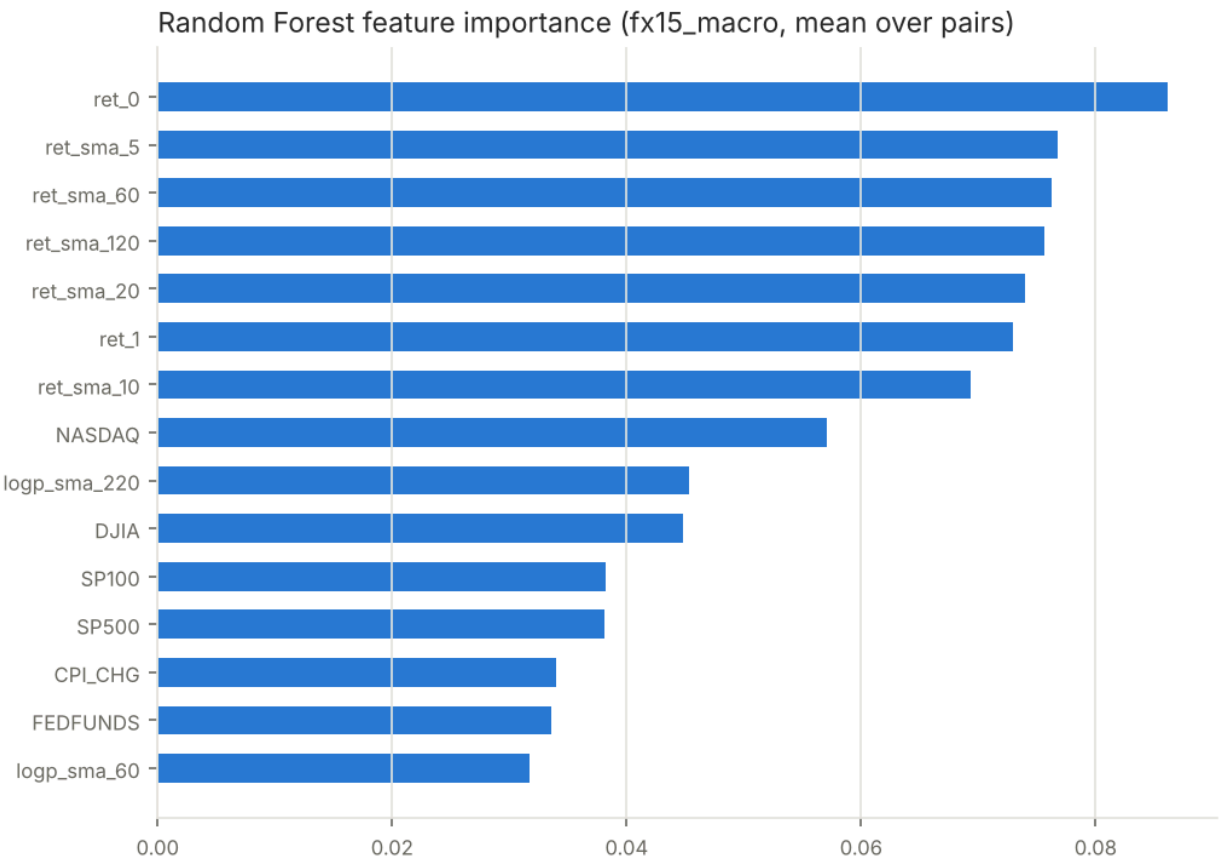


Figure 6. Mean impurity importance over the three pairs (`fx15_macro`).

The top features are the recent-return features (today's return and its 5–120 day averages), then Nasdaq and other index returns. CPI, Fed funds and debt rank low. High importance does not mean useful prediction: a deep forest splits on whatever varies most, including noise. That is consistent with its negative test R^2.

9. Discussion

9.1 Direction prediction is a coin flip, in our data and in the paper's. Our Table 2 sits in the same 0.50–0.53 band as the paper's Table 1, and no model beats chance once multiple comparisons are accounted for. The paper's interpretation agrees ("accuracy values are all around 50%").

9.2 The paper's regression results do not replicate. Their Random Forest R^2 of 0.70–0.78 on daily log returns is incompatible with their own ~0.5 sign accuracy. A model explaining three quarters of the variance of returns would get the direction right most of the time. On public data, the same model, features and protocol give test R^2 of −0.06 to −0.02. The paper gives the R^2 formula but not exactly which predictions it was computed on. Likely explanations are R^2 computed on in-sample or overlapping data, or features that include information from day t+1.

9.3 The random walk wins. The validation set chooses a Lasso so heavily regularised that it predicts a constant. In effect, the data are telling us that the best forecast of tomorrow's return is (almost) zero. That is Meese & Rogoff's finding, now from a machine-learning pipeline. Momentum (assuming today's move repeats) is much worse than the random walk (R^2 vs RW = −1.11), which confirms daily moves have no simple persistence.

9.4 Robustness to data cleaning. We also ran the full pipeline on the raw data, before removing the 9 bad prices.

- **Unchanged:** Lasso is still selected. Every Exp 1 accuracy stays in 0.49–0.53. Random Forest R^2 vs RW on `fx15_macro` is still negative (−0.060 raw vs −0.034 cleaned).
- **Changed:** Random Forest's no-cost profit on EUR moves from $104.32 to $100.75, showing how fragile the backtest numbers are.

9.5 Did the economic factors help? Not reliably. Adding them raised some Exp 1 accuracies (MLP 0.522 → 0.529, SVM 0.503 → 0.518) and lowered others (Lasso 0.520 → 0.505). In Exp 2 they made most regressors slightly worse (e.g. MLP −0.014 → −0.031). With monthly or quarterly updates and publication lags, they carry almost no information about tomorrow.

9.6 Why the backtest misleads. A single ~17-month path with no costs can show gains by luck. During development, models run on synthetic random-walk data, which is unpredictable by construction, still ended at $110–$120 at zero cost. The paper's $110.50 is within that range. Costs, a benchmark and a significance test are needed before reading anything into final cash.

10. Limitations and Future Work

- **Single split.** One 17-month test window. Walk-forward evaluation over many windows would give a distribution of results instead of one number.
- **Significance testing.** Our binomial tests assume independent days and cover direction only. A Diebold–Mariano test on forecast errors versus the random walk is the standard stronger check.
- **Data.** Yahoo instead of Bloomberg (shorter history, bad prices that needed cleaning, August 2008 missing for EUR/JPY). The NBER recession flag is declared months after the fact, so even lagged it contains hindsight.
- **Costs.** A flat 2 bp per side, with no bid–ask modelling and no interest earned on the currency held.
- **Future work:** walk-forward validation; Diebold–Mariano tests; LSTM/GRU models (the paper's own suggestion); higher-frequency data or interest-rate differentials, which economic theory links to currency moves.

11. Conclusion

We rebuilt CS229 #34 end to end on public data and kept everything the paper specified. The sign-prediction results replicate: about 50%, no better than a coin flip. The headline regression and trading results do not. Random Forest's test R^2 is negative rather than 0.70–0.78, the validation set prefers a constant forecast, and every trading combination loses money once a 2 bp cost is included. The two results that briefly looked like skill disappeared on the validation period and under different random seeds. Our main lesson is methodological: **without a random-walk baseline, point-in-time data and validation checks, daily FX forecasting results are easily overstated.** With them, the 1983 conclusion still stands.

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Appendix

A. Reproduce. `pip install -r requirements.txt`, then `python main.py` (~10 min on a laptop), `python -m src.diagnose`, and `python -m src.diagnose MLP fx15_macro --seeds`.

B. Full result tables. `results/exp1_sign_by_pair.csv`, `results/exp2_return_by_pair.csv`, `results/backtest_summary.csv`, `results/significance_exp1.csv`, `results/significance_exp2_direction.csv`.

C. Contributions. Catherine D (CS120): data collection (Yahoo, FRED), cleaning and bad-print analysis, feature engineering and point-in-time alignment, exploratory analysis, report §1–5. Yashwitha (CS121): models and validation-based selection, metrics, profit simulation, significance and robustness checks, report §6–9. Both: research design, discussion, conclusion, presentation.